

A 33 yo M with a history of a kidney transplant (CMV donor +/- recipient) 6 months ago for IgA nephropathy is found to have a Cr of 3.4 (prior Cr 1.5). He is on tacrolimus, prednisone, and mycophenolate.

Click here

What is your differential diagnosis?

Note for teachers: Remind your learners that typical causes of AKI still apply in patients with a kidney transplant. Quickly, review the non-transplant causes, but do not focus on the differential for them. Check out [our talk on the differential for AKI](#).

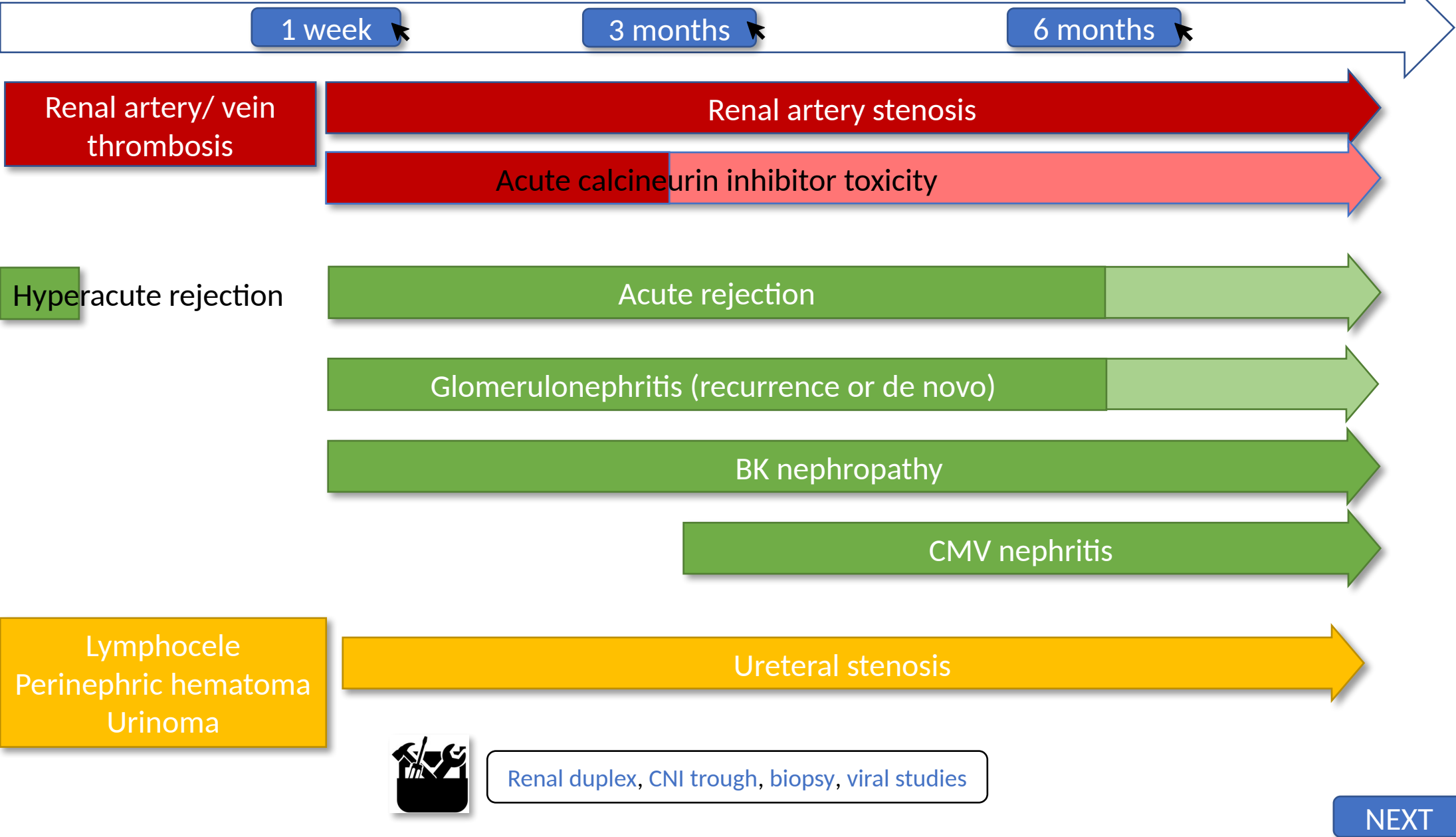
- Etiologies can be classified by:
- 1) Pre-, intra-, and post-renal causes
 - 2) [Timing from transplant](#)

Click here

	PRE-RENAL	INTRA-RENAL	POST-RENAL
Typical/ Non-transplant specific	- Sepsis - Medications (NSAIDS, - Differential? - Hepatorenal - Volume depletion	- Acute tubular necrosis - Contrast nephropathy - Differential? - nephritis	- Obstruction from 3C's (Calculi, Clot, Cancer) - Differential?
Transplant specific	- Acute calcineurin - Differential? - thrombosis	- Acute rejection - Differential? - Chronic CNI toxicity - Pyelonephritis	- Ureteral stenosis - Differential? - hematoma, urinoma

Pre-renal, intra-renal, and post-renal etiologies can be further divided into transplant-specific and typical/usual causes.

How does the time from transplant affect our differential diagnosis? (Click on the timeline buttons or "pre-renal, intra-renal, post-renal" to show the differential. Click on the arrows to display additional information. Click again to make the information disappear.)





A 33 yo M with a history of a kidney transplant (CMV donor +/- recipient +) 6 months ago for IgA nephropathy is found to have a Cr of 3.4 (prior Cr 1.5). He is on tacrolimus, prednisone, and mycophenolate.

Case 1

ROS: new numbness/ tingling in his hands/feet, pain and redness in his L 1st metatarsal.

Meds: fluconazole 2 weeks ago for a yeast infection. No other med changes.

Exam: BP 160/ 92, other VS normal. No tenderness over his graft.

135	52	
6.9	3.4	

FeNa - 0.7%

UA - clear with hyaline casts, no protein, WBCs, or RBCs



Cr improved to 3.1 with 1 L IVF.

CNI trough elevated at 14, goal 6-8.

Normal arterial flow, no hydronephrosis.

Not done.

Urine BK PCR level elevated at 14k.

Acute CNI toxicity resulting in neurotoxicity, gout flare, ↑ BP, ↑ K. Has underlying BK nephropathy. Fluconazole interacts with CNIs to ↑ CNI levels.

Case 2

ROS: fevers/chills for 2 days, abdominal pain, dysuria, nausea/vomiting.

Exam: T 39.4C, HR 115, BP 160/ 92. Tenderness over his graft.

135	32	
3.5	3.4	

FeNa - 1.2%
Urine Pr:Cr - 2.6

UA - cloudy, + leukocyte esterase, + nitrite + WBCs, + RBCs, + protein, + bacteria.



Cr improved to 3.1 with 1 L IVF.

CNI trough within range at 7.

Normal arterial flow, no hydronephrosis.
Perinephric stranding.

Acute cellular rejection. No DSAs.

Urine BK/adenovirus PCR negative. CMV PCR negative.

Acute cellular rejection and sepsis from acute graft pyelonephritis.

Case 3

Renal biopsy yesterday for uprending Cr, complicated by hematuria.

ROS: acute onset suprapubic/abdominal pain, no urine output for 12 hours

Exam: VSS. Abdomen tender over graft/suprapubic region.

135	32	
3.5	3.4	

FeNa - 5.7%
Urine Pr:Cr - 0.6

UA - none collected



Cr unchanged at 3.3 after 1 L IVF.

CNI trough within range at 7.

Normal arterial flow, graft hydronephrosis.
Bladder with 700 cc of urine and retained clot.

Glomerulosclerosis and interstitial fibrosis concerning for chronic CNI toxicity.

Not sent.

Acute clot retention resulting in obstructive uropathy on background of chronic CNI toxicity.